

CompTIA Security+

SY0-701 — Detailed Domain Notes

with Reasoning & Application

Every concept in three layers — **What it is**, **Why it works**, and **How it is applied** — grounded in the SY0-701 blueprint and tagged with objective codes and confidence levels. Built for exam readiness and operational understanding, not memorization.

Exam at a Glance

Exam Attribute	Value
Exam code	SY0-701 (V7), objectives v5.0
Questions / Time	Up to 90 questions in 90 minutes
Question types	Multiple-choice (single & multi) + Performance-Based Questions (PBQs)
Passing score	750 on a 100–900 scale
Renewal	3 years (CompTIA CE program)

Domain Weightings (weight first)

#	Domain	Weight	Study priority
4	Security Operations	28%	Highest — most PBQs cluster here
2	Threats, Vulnerabilities & Mitigations	22%	High
5	Security Program Management & Oversight	20%	High — memorization-friendly GRC
3	Security Architecture	18%	Medium — device-placement PBQs
1	General Security Concepts	12%	Foundational vocabulary

How to Use These Notes

These notes are structured around **causal chains**, not flat definitions. Each concept is deconstructed so you learn *why* a control works and *when* it fails — the reasoning the exam tests in its scenario and performance-based questions.

The three-layer reading model

Layer	What it answers	Why it matters for the exam
What	The precise definition and its component parts.	Recall-tier questions and vocabulary.
Why it works	The causal mechanism — inputs → mechanism → outcome.	Application-tier "given a scenario" questions.
Applied	A realistic deployment or attack/defense scenario.	PBQs and analysis-tier reasoning.

Where a concept has a critical failure mode, a red **Fails when** line marks the conditions under which the control breaks — the invalidation logic behind analysis-tier questions.

Labels you will see

Label	Meaning
[1.1] ... [5.6]	Official SY0-701 sub-objective code the content maps to.
Recall / Application / Analysis	Cognitive tier the exam uses to test the concept.
Confidence: HIGH / MODERATE	Evidence weight behind the claim. HIGH = settled standard; MODERATE = context-dependent or version-sensitive.
[VERSION-SENSITIVE]	Feature or best practice that shifts with vendor/standard versions — verify currency.

Study sequence. Work top-down by exam weight: Domain 4 first (28%), then 2, 5, 3, and 1. Domain 5 (GRC) is memorization-friendly but decides passes — do not defer it. Treat every "Applied" block as a mini-PBQ: cover the answer, reason through the scenario, then check.

DOMAIN 1 — General Security Concepts

12%

Foundational vocabulary and models that every later domain builds on: what controls exist and why they are categorized as they are, the goals security serves (CIA, AAA, non-repudiation), the zero-trust model, the security value of change management, and the cryptographic primitives that enforce it all.

1.1 — Security Controls

A security control is any measure that reduces risk. The exam tests two independent axes: the **category** (what kind of mechanism) and the **type** (what effect it has on the threat timeline). A single control has one category but can serve multiple types.

Category	It is...	Example
Technical	Enforced by technology/system logic.	Firewall, encryption, access-control list, EDR.
Managerial	Administrative — policy and process direction.	Risk assessments, security policy, SDLC reviews.
Operational	Executed by people day-to-day.	Guard shifts, awareness training, backup routines.
Physical	Protects the tangible environment.	Locks, fences, bollards, badge readers, mantraps.

Type	Function	Example
Preventive	Stops an incident before it happens.	Firewall rule, least privilege, door lock.
Deterrent	Discourages the attacker from trying.	Warning banner, visible cameras, signage.
Detective	Identifies an incident in progress/after.	IDS, SIEM alert, log review, motion sensor.
Corrective	Restores after an incident.	Backup restore, patching, IPS quarantine.
Compensating	Alternate control when primary is infeasible.	MFA + monitoring where a legacy app can't be patched.
Directive	Directs/instructs behavior by policy.	Acceptable use policy, "authorized personnel only" sign.

Category vs. type reasoning [1.1] · Recall→Analysis

What. Category = the nature of the mechanism; Type = its effect relative to the attack timeline (before / during / after).

Why it works. They are orthogonal because the same mechanism can act at different points in time. A camera is a *physical* category control, yet it is *deterrent* (if visible) and *detective* (if recorded) at once. The exam exploits this to test whether you conflate the two axes.

Applied. Control-matching PBQ: "an encrypted backup that rebuilds a ransomed server" → category technical, type corrective. "A banner threatening prosecution" → managerial/directive, type deterrent. Classify on both axes independently.

Confidence: HIGH — settled CompTIA taxonomy.

1.2 — Fundamental Concepts: CIA, AAA & Zero Trust

CIA Triad [1.2] · Recall→Application

What. The three goals of information security. **Confidentiality** = no unauthorized disclosure; **Integrity** = data stays accurate/unaltered except by authorized change; **Availability** = authorized users can reach resources when needed.

Why it works. These three are in tension, which is why the triad is a design framework, not a checklist. Encrypting everything (confidentiality) can slow access (availability); locking data read-only (integrity) can block legitimate edits. Every control decision trades along these axes — naming the axis a control serves is the reasoning the exam wants.

Applied. Map an attack to the leg it violates: ransomware encrypting your files = availability (and integrity); a database leak = confidentiality; a tampered wire-transfer amount = integrity. The mitigation you pick should restore the specific leg that was hit.

Confidence: HIGH.

Non-repudiation & the AAA framework [1.2] · Recall→Application

What. Non-repudiation guarantees an actor cannot deny an action (via digital signatures/logs). **AAA** = Authentication (prove identity), Authorization (grant appropriate access), Accounting (log activity).

Why it works. The order is causal and non-negotiable: you cannot authorize what you have not authenticated, and you cannot hold anyone accountable without accounting records. Non-repudiation depends on both authentication (who) and accounting (what/when) being trustworthy — a signed, logged action is one the actor cannot repudiate.

Applied. A user logs in with MFA (authN), is granted read-only to a share by role (authZ), and every file access is logged (accounting). If they later delete a record, the signed audit trail provides non-repudiation for the investigation.

Confidence: HIGH.

Zero Trust — control plane vs. data plane [1.2] · Application→Analysis

What. Zero trust assumes no implicit trust by location or prior authentication; every request is verified. It splits into a **control plane** (policy engine + policy administrator that make and push the trust decision) and a **data plane** (the policy enforcement point that actually allows/blocks the traffic).

Why it works. The perimeter model failed because once inside, an attacker moved laterally unchecked. Zero trust removes "inside" as a trust signal: the control plane re-evaluates identity, device posture, and context on every request, and the data plane enforces the verdict at the resource. Separating decision from enforcement is what lets policy scale independently of where the resource sits.

Applied. Adaptive access: a user authenticated an hour ago requests a sensitive DB. The policy engine re-checks device compliance and risk score; if the device just failed a posture check, the policy administrator revokes the session and the enforcement point (data plane) drops the connection — no lateral free pass.

Confidence: HIGH — NIST SP 800-207 aligned.

1.3 — Change Management & Its Security Impact

Change management process [1.3] · Recall→Application

What. A structured process to move systems from a current to a desired state with approval, assessment, and documentation. Core elements: a **Change Advisory Board (CAB)** approves; a **change owner** advocates; **stakeholders** are consulted; an **impact analysis** precedes action; a **maintenance window**, **backout plan**, and **testing** bound the execution.

Why it works. Unmanaged change is a top cause of outages and security gaps because changes create side effects — a firewall rule edit can silently open a path; a patch can break a dependency. The process forces the side effects to

be predicted (impact analysis), reversible (backout plan), and time-boxed (window) before they reach production.

Applied. A team must patch a public web server. Change management requires: CAB approval, an impact analysis flagging a dependent legacy app, a Saturday maintenance window, a tested backout snapshot, and updated diagrams/tickets afterward. Skipping the backout plan is the classic exam "what was missing?" answer.

Fails when changes bypass the CAB, allow/deny lists aren't updated with the change, or documentation/diagrams drift — leaving responders with a false model of the network during the next incident.

Confidence: HIGH.

Technical implications tested: **allow/deny list** updates, **restricted activities**, **downtime** and service/application **restarts**, **legacy application** and **dependency** effects, and **version control** of configuration and documentation.

1.4 — Cryptographic Solutions

Cryptography is the technical engine behind confidentiality, integrity, authentication, and non-repudiation. Know which primitive delivers which CIA/AAA property — the exam maps them constantly.

Primitive	Delivers	Key facts
Symmetric encryption	Confidentiality	One shared key; fast; scales poorly (n^2 keys). AES-256.
Asymmetric encryption	Confidentiality + key exchange	Public/private key pair; slow; solves key distribution. RSA, ECC.
Hashing	Integrity	One-way, fixed-length digest; no key. SHA-256. Collisions break it.
Digital signature	Integrity + authentication + non-repudiation	Hash of message encrypted with sender's private key.
Salting	Integrity of stored secrets	Random value added before hashing to defeat rainbow tables.
PKI / certificates	Authentication + trust	CA binds a public key to an identity via a signed X.509 cert.

Why asymmetric + symmetric are used together [1.4] · Application→Analysis

What. Real protocols (TLS) use asymmetric crypto to exchange a key, then switch to symmetric for the bulk data — a hybrid cryptosystem.

Why it works. Each primitive fixes the other's weakness. Symmetric is fast but cannot safely distribute its key over an untrusted network. Asymmetric solves key distribution but is computationally expensive for large data. So the expensive asymmetric step runs once to establish a shared symmetric session key, and the cheap symmetric cipher carries the session. Speed and safe key exchange are both satisfied.

Applied. TLS handshake: the client uses the server's public key (from its certificate) to agree a session key, then all page data is AES-encrypted with that session key. A digital signature on the certificate lets the client verify the server's identity (authentication) before trusting the key.

Fails when the CA is compromised or a certificate isn't validated — the client trusts an attacker's public key and the entire chain of confidentiality and authentication collapses.

Confidence: HIGH.

DOMAIN 2 — Threats, Vulnerabilities & Mitigations

22%

Who attacks, how they get in, what weaknesses they exploit, how you detect them, and how you shut them down. This domain is scenario-heavy — expect to map an indicator to an attack and an attack to a mitigation.

2.1 — Threat Actors & Motivations

Actor	Sophistication / resources	Typical motivation
Nation-state / APT	Very high; funded, persistent	Espionage, strategic disruption, IP theft.
Organized crime	High; well-funded	Financial gain (ransomware, fraud).
Hacktivist	Moderate; ideological	Political/social message, disruption, doxxing.
Insider threat	Variable; already trusted	Revenge, financial, or unintentional error.
Unskilled attacker ("script kiddie")	Low; uses others' tools	Notoriety, curiosity, opportunism.
Shadow IT	N/A; internal risk	Convenience — unsanctioned tools create exposure.

Attributes drive the defense [2.1] · Application

What. Actors differ on internal vs. external origin, resources/funding, and sophistication.

Why it works. These attributes are causal, not trivia: they predict persistence and technique, which dictates the proportionate control. A nation-state's funding means it can burn zero-days and wait months, so defense emphasizes detection depth and threat hunting. An unskilled attacker relies on known exploits, so patching and basic hygiene stop most of them. Matching control effort to actor capability is the reasoning tested.

Applied. Scenario: repeated, slow, targeted logins from varied infrastructure against one executive = likely APT → respond with enhanced monitoring and hunting, not just a password reset. A noisy mass-exploit scan of a known CVE = commodity actor → prioritize the patch.

Confidence: HIGH.

2.2 — Threat Vectors, Attack Surfaces & Social Engineering

Technique	Vector	Tell / defense
Phishing	Email	Urgency + link; defense: filtering, DMARC/DKIM/SPF, training.
Spear phishing / whaling	Targeted email	Personalized; targets execs; verify out-of-band.
Smishing	SMS	Malicious link/number; don't tap, verify sender.
Vishing	Voice call	Pretext for info; callback via known number.
BEC	Email impersonation	Fake exec wire request; dual-approval process.
Pretexting	Any	Fabricated scenario to build trust.
Watering hole	Compromised site	Infects a site the target group frequents.

Why social engineering beats technical controls [2.2] · Application→Analysis

What. Threat vectors are the paths in (message, image, file, voice, removable media, unsecure networks, open ports, supply chain); the attack surface is the sum of all such paths.

Why it works. Social engineering targets the human, who is outside the trust boundary your technical controls enforce. No firewall rule stops a user who is manipulated into voluntarily clicking, calling back, or approving a wire — the attacker borrows the user's legitimate access. That is why awareness training (a control) is a genuine security control, not a soft add-on: it hardens the one attack surface technology can't patch.

Applied. BEC scenario: an "urgent" email from the "CFO" asks finance to wire funds before end of day. The technical controls pass (the mailbox may be spoofed or look-alike domain). The mitigation is procedural — out-of-band verification and dual approval — because the exploit is trust, not code.

Confidence: HIGH.

2.3 — Vulnerability Types

Vulnerability	Where	Mechanism / risk
Buffer overflow	Application memory	Writes past a buffer boundary to overwrite adjacent memory / hijack execution.
Race condition (TOCTOU)	Concurrent processes	Timing gap between check and use lets state change maliciously.
Memory injection	Application	Attacker code inserted into a running process's memory.
SQL injection	Web / database	Unsanitized input alters a SQL query to read/modify data.
XSS	Web (client)	Malicious script runs in a victim's browser on a trusted site.
VM escape	Virtualization	Guest breaks isolation to reach the hypervisor/host.
Misconfiguration	Any	Default creds, open S3 bucket, excessive permissions.
Zero-day	Any	Exploited before a patch exists; no signature yet.

Buffer overflow — the mechanism [2.3] · Analysis

What. A process writes more data into a fixed-size buffer than it allocated; the excess overwrites adjacent memory, including the return address on the stack ("smashing the stack").

Why it works. Memory has no built-in bounds checking in low-level languages, so an overlong input flows into whatever sits next in memory. If the attacker overwrites the saved return address with a pointer to their own injected code (often reached via a NOP slide), the CPU jumps there when the function returns — arbitrary code execution. The root cause is missing input validation, which is why the mitigation lives at the code and OS layer.

Applied. Mitigations map to the mechanism: input validation stops the overlong write; ASLR randomizes memory addresses so the attacker can't reliably locate their payload; DEP/NX marks the stack non-executable so injected code won't run even if placed. On the exam, "which control defeats a buffer overflow" → ASLR/DEP + input validation.

Confidence: HIGH.

SQL injection vs. XSS — who gets hurt [2.3] · Application→Analysis

What. SQLi injects SQL into a query to attack the **server's database**; XSS injects script that runs in another **user's browser** on a trusted site.

Why it works. Both stem from the same root — untrusted input treated as trusted code — but at different execution contexts. SQLi succeeds because input is concatenated into a query the database engine executes; XSS succeeds because input is echoed into a page the browser executes. Identifying the execution context tells you both the victim and the fix: parameterized queries for SQLi, output encoding + CSP for XSS.

Applied. Log/PBQ tell: a web log request containing ' OR 1=1 -- in a parameter = SQLi attempt. A request stuffing <script> into a comment field that later renders for other users = stored XSS.

Confidence: HIGH.

2.4 — Indicators of Malicious Activity

Scenario/PBQ-heavy: map an observed indicator to the attack. Know malware types and the signature network/application/cryptographic attacks.

Malware	Behavior / indicator
Ransomware	Mass file encryption, ransom note, spike in file writes.
Trojan	Legitimate-looking program with hidden malicious function.
Worm	Self-propagates across the network without user action.
Rootkit	Hides at OS/kernel level; evades AV; persistence.
Keylogger	Captures keystrokes; credential theft indicator.
Logic bomb	Dormant code triggered by a condition/date.
Spyware	Covert data collection / exfiltration.

Attack	Indicator you'd see
DDoS	Traffic flood from many sources; service unavailable; bandwidth saturation.
DNS attack (poisoning/hijack)	Users routed to wrong IPs; unexpected DNS record changes.
On-path (MITM)	Certificate warnings; ARP table anomalies; session data intercepted.
Credential replay	Reused captured token/hash; login from anomalous time/location.
Password spraying	One common password tried across many accounts; low-and-slow lockouts.
Privilege escalation	A standard account suddenly performing admin actions.

Reading indicators as a causal chain [2.4] · Analysis

What. An indicator of compromise (IoC) is an observable artifact — a log entry, traffic pattern, file change — that signals an attack.

Why it works. Each attack leaves a characteristic footprint because of how it works. DDoS saturates bandwidth, so the indicator is volumetric; password spraying avoids lockouts by trying one password widely, so the indicator is many accounts with a single failed attempt rather than one account with many. Reasoning backward from the mechanism to the expected artifact is exactly the analysis the PBQ log-reading questions test.

Applied. PBQ: authentication log shows one failed login each for 200 accounts within 60 seconds, same source. That low-and-slow-across-accounts pattern = password spraying (not brute force, which hammers one account). Mitigation: MFA + anomalous-login detection, not just account lockout.

Confidence: HIGH.

2.5 — Mitigation Techniques

Layered mitigation (defense in depth) [2.5] · Application

What. Core techniques: **segmentation, access control, isolation, patching, encryption, monitoring, least privilege, configuration enforcement, decommissioning, and hardening.**

Why it works. No single control is complete, so mitigations are layered to limit blast radius. Segmentation contains lateral movement; least privilege limits what a compromised account can do; patching removes the vulnerability; monitoring catches what prevention misses. The layers are chosen so a failure of one is caught or contained by another — that redundancy is the whole point of defense in depth.

Applied. Ransomware response mapping: segmentation stops it spreading to other VLANs; least privilege limits which shares it can encrypt; offline backups (corrective) enable recovery; EDR monitoring detects the mass-write behavior early. The exam wants the mitigation that matches the specific stage of the attack.

Fails when layers are redundant in kind but not independent — e.g. two detective controls and no containment. Diversity across control types matters more than count.

Confidence: HIGH.

DOMAIN 3 — Security Architecture

18%

How the shape of the environment changes its risk: cloud and virtualization models, where to place defensive devices, how to protect data in every state, and how to keep systems available through failure. Home of the device-placement drag-and-drop PBQ.

3.1 — Architecture Models

Model	Security implication
Cloud (shared responsibility)	Provider secures the cloud; customer secures what's in it. Line shifts by IaaS/PaaS/SaaS.
Virtualization	Consolidation + isolation; risk = VM escape, hypervisor as single point of failure.
Containerization	Lightweight, shared kernel; weaker isolation than VMs; image supply-chain risk.
Serverless	No OS to patch (provider's job); risk shifts to code, config, and function permissions.
Microservices	Many small services; larger attack surface / API security; better fault isolation.
IaC	Config as code — repeatable & auditable; a bad template propagates the flaw everywhere.
SDN	Centralized control plane; agile segmentation; controller compromise is high impact.
IoT / ICS-SCADA / RTOS	Long-lived, hard to patch, availability-first; segment aggressively.

Shared responsibility — why the line moves [3.1] · Application→Analysis

What. In cloud, security duties split between provider and customer, and the split depends on the service model.

Why it works. The more the provider manages, the more security they own — but only up to the layer they control. In IaaS the customer patches the OS and app; in PaaS the provider handles the OS but the customer still owns code and data; in SaaS the provider handles almost everything except data classification, access, and configuration. The causal rule: you always own your data and identities regardless of model. Misreading this line is the top cloud-breach root cause (e.g., public storage buckets).

Applied. Scenario: a company on SaaS suffers a data leak from a misconfigured sharing setting. Whose fault? The customer's — configuration and access are always the customer's responsibility even in SaaS. The provider's infrastructure was never breached.

Confidence: HIGH.

3.2 — Secure Enterprise Infrastructure & Device Placement

Device	Purpose	Placement logic
Firewall	Filter traffic by rule	Boundary between zones (edge, between DMZ and internal).
Screened subnet (DMZ)	Host public services	Between two firewalls; isolates internet-facing servers.
IDS / IPS	Detect / block by signature	IDS out-of-band (span port); IPS inline in the traffic path.
Jump server	Controlled admin access	In a management segment; single audited entry to sensitive hosts.

Device	Purpose	Placement logic
Proxy	Mediate & inspect requests	Between clients and internet (forward) or before servers (reverse).
Load balancer	Distribute + availability	In front of a server pool; also TLS termination.
VPN / SASE	Secure remote access	Concentrator at edge; SASE delivers it as cloud service.

IDS vs. IPS placement — why it matters [3.2] · Analysis

What. An IDS detects and alerts; an IPS detects and blocks. Placement follows the function.

Why it works. An IPS must sit *inline* — every packet passes through it — because it has to be able to drop malicious traffic in real time; that also makes it a potential bottleneck and single point of failure. An IDS sits *out-of-band* on a mirror/span port because it only needs a copy to analyze; it can't block, but it can't break traffic flow either. The placement is a direct consequence of "must act" vs. "only observe."

Applied. Device-placement PBQ: asked to stop an active exploit at the perimeter, place an IPS inline at the edge. Asked to monitor internal traffic for anomalies without risking latency, place an IDS on a span port. Jump server goes in the management zone as the sole path to admin the internal servers.

Confidence: *HIGH*.

3.3 — Data Protection Strategies

Protect data by its **state** and its **value**. Three states: *at rest (stored)*, *in transit (moving)*, *in use (being processed)*. Classification drives how much protection each gets.

Technique	What it does	Best use
Encryption	Renders data unreadable without a key	At rest (disk) and in transit (TLS).
Hashing	One-way integrity check	Verify data/file wasn't altered; store password digests.
Masking	Hides part of a value (****1234)	Display contexts where full value isn't needed.
Tokenization	Swaps data for a non-sensitive token	PCI card data; token maps back only in a secure vault.
Obfuscation	Makes data harder to interpret	Lower-assurance concealment; not a strong control alone.
Segmentation	Separates data by sensitivity	Keep regulated data in an isolated, tightly controlled zone.

Tokenization vs. encryption — why choose one [3.3] · Analysis

What. Encryption transforms data with a reversible algorithm + key; tokenization replaces it with a random surrogate that has no mathematical relationship to the original.

Why it works. Because a token is not derived from the data, there is nothing to decrypt if the token store is breached — the sensitive value never lives in the processing system at all. Encryption keeps the data present but scrambled, so key compromise reveals everything. For narrowing regulatory scope (PCI DSS), removing the data entirely (tokenization) beats protecting data you still hold (encryption). The choice follows whether you need the real value in place.

Applied. A merchant tokenizes card numbers so its app database stores only tokens; the real PAN sits in the processor's vault. A stolen app DB yields useless tokens, shrinking PCI audit scope. Disk encryption alone wouldn't reduce scope because the plaintext still passes through the app.

Confidence: HIGH.

Also tested: **data sovereignty** (data is subject to the laws of the country it physically resides in) and **permission restrictions** aligned to classification.

3.4 — Resilience & Recovery

Concept	Meaning
RTO (Recovery Time Objective)	Max tolerable time to restore a service after outage.
RPO (Recovery Point Objective)	Max tolerable data loss, measured in time (drives backup frequency).
Hot site	Fully equipped, running — fail over in minutes; most costly.
Warm site	Partially equipped — hours to bring up; moderate cost.
Cold site	Space + power only — days to stand up; cheapest.
High availability	Redundancy (clustering, load balancing) to avoid downtime.
Platform diversity / multi-cloud	Avoid a shared flaw or single vendor failure.

RTO vs. RPO drive different investments [3.4] · Application→Analysis

What. RTO is about *time to recover*; RPO is about *how much data you can lose*. They are independent knobs.

Why it works. RTO is bought with recovery infrastructure — a hot site and clustering shrink downtime. RPO is bought with backup/replication frequency — hourly snapshots shrink data loss to an hour. Confusing them leads to spending on the wrong control: a hot site with nightly backups still loses a day of data (bad RPO) even though it recovers fast (good RTO). Matching the metric to the control is the reasoning tested.

Applied. Scenario: a trading system tolerates 5 minutes of downtime but zero data loss → invest in synchronous replication (RPO≈0) plus a hot site (low RTO). A monthly report server tolerates a day down and a day of loss → a cold site and daily backups suffice, at far lower cost.

Confidence: HIGH.

The largest domain and where most PBQs cluster: hardening and secure baselines, asset lifecycle, vulnerability management, monitoring tooling, hands-on capability changes (firewall rules, email security), identity and access management, automation, incident response, and log-driven investigation.

4.1 — Secure Computing Techniques & Hardening

Secure baselines & hardening [4.1] · Application

What. A **secure baseline** is a known-good, minimally-configured standard for a system; **hardening** is reducing attack surface by disabling unneeded services, closing ports, enforcing config, and patching. Includes wireless (WPA3 + RADIUS), MDM for mobile (BYOD/COPE/CYOD), and sandboxing.

Why it works. Every enabled feature, open port, and default account is a potential entry point. Hardening works by subtraction — the smaller the attack surface, the fewer paths an attacker can take, and the fewer things you must monitor and patch. A baseline makes "secure" repeatable and measurable: you can detect drift because you know the intended state.

Applied. New server build: apply the CIS/secure baseline image, disable Telnet/SMBv1, remove default creds, enable the host firewall, enroll in patch management. WPA3 replaces WPA2 because SAE resists offline dictionary attacks on the handshake — the mechanism, not just "newer = better."

Confidence: HIGH.

Model	Ownership	Trade-off
BYOD	Employee-owned	Cheapest hardware; weakest control; hidden security cost.
COPE	Corp-owned, personal use OK	Strong MDM control; higher cost.
CYOD	Employee picks from approved list	Balance of choice and control.

4.2 — Asset Management

Asset lifecycle & secure disposal [4.2] · Application

What. Manage assets from acquisition → assignment/accounting (clear ownership + classification) → monitoring → secure disposal/decommissioning. Disposal follows **NIST SP 800-88**: sanitize, destroy, and certify.

Why it works. You can only protect what you know you have — an unaccounted asset is unpatched, unmonitored, and a blind spot. Ownership assignment creates accountability; classification decides protection level. At end of life, data outlives the device's usefulness, so sanitization (or physical destruction for high-classification media) prevents data remanence — recoverable data on discarded drives.

Applied. Decommission a server holding regulated data: sanitize per 800-88 (crypto-erase or purge), or physically destroy (shred/degauss) for Secret/Top Secret, then obtain a certificate of destruction for the audit trail. Selling drives after a quick format is the classic wrong answer — formatting doesn't remove data.

Fails when decommissioned assets skip sanitization or aren't removed from the CMDB — leaving recoverable data and phantom inventory entries that corrupt vulnerability reporting.

Confidence: HIGH.

4.3 — Vulnerability Management

Identify → prioritize → remediate → validate [4.3] · Application→Analysis

What. The cycle: discover vulnerabilities (scans, threat feeds, pen tests, bug bounties), prioritize (**CVSS** severity, **CVE** identifier, exploitability/EPSS, asset value), remediate (patch or compensating control), then **rescan/validate** and report.

Why it works. You can never fix everything at once, so prioritization is the core skill. CVSS gives a severity (0–10) but severity alone misleads — a 9.8 on an isolated test box matters less than a 7.5 on an internet-facing crown-jewel system. Combining CVSS with exploit likelihood and asset context ranks real risk. Rescanning closes the loop: without validation you don't know the fix worked.

Applied. A scan returns 400 findings. Sort by CVSS, re-rank by internet exposure and business criticality, patch the top tier, apply compensating controls (segmentation, WAF) where no patch exists, and rescan to confirm. A CVSS 10 with no known exploit on an air-gapped host waits behind a CVSS 8 actively exploited on the DMZ.

Fails when teams patch by raw CVSS alone, ignoring exposure and exploit activity — burning effort on unreachable bugs while an exploited medium sits on the perimeter.

Confidence: HIGH.

4.4 — Alerting & Monitoring Tools

Tool	Purpose (match tool → job)
SIEM	Aggregates and correlates logs from many sources; central alerting/analysis.
SCAP	Standardizes security config/vuln checks so tools interoperate.
DLP	Detects/blocks sensitive data leaving the org (email, USB, cloud).
Antivirus / EDR	Signature + behavioral endpoint threat detection & response.
SNMP traps	Device-initiated alerts on state changes/thresholds.
NetFlow	Traffic-flow metadata for volume/pattern analysis.
Vulnerability scanner	Finds missing patches/misconfigurations across hosts.

Why the SIEM is the correlation engine [4.4] · Application→Analysis

What. A SIEM ingests logs from firewalls, endpoints, servers, and applications, normalizes them, and correlates across sources to raise alerts.

Why it works. A single log source only sees its own slice. An attack spans layers — a failed VPN login, then a firewall allow, then a privilege change — and each event alone looks benign. Correlation turns scattered, individually-innocent events into one detectable narrative. That cross-source correlation is why the SIEM, not any single sensor, is where detection logic lives.

Applied. Tool-to-purpose PBQ: "block a spreadsheet of SSNs from being emailed out" → DLP. "Alert when a router interface goes down" → SNMP trap. "See which internal host is beaconing large volumes outbound" → NetFlow. "Correlate auth failures with a later admin action" → SIEM.

Confidence: HIGH.

4.5 — Modify Enterprise Capabilities

Hands-on capability changes — home of the firewall-rule-ordering PBQ. Know email authentication (SPF/DKIM/DMARC), filtering, and endpoint response tooling.

Mechanism	What it verifies
SPF	Which mail servers are authorized to send for a domain (DNS TXT record).
DKIM	A cryptographic signature proving the message wasn't altered and came from the domain.
DMARC	Policy telling receivers what to do when SPF/DKIM fail (none/quarantine/reject) + reporting.
DNS filtering	Blocks resolution of known-malicious domains.
NAC	Admits devices to the network only if they meet posture policy.
EDR / XDR	Endpoint (EDR) or cross-layer (XDR) detection and automated response.
FIM	File Integrity Monitoring — alerts on unauthorized changes to critical files.

Firewall rule ordering — why order is everything [4.5] · Analysis

What. Firewalls evaluate rules **top-down** and act on the **first match**. Most rule sets end with an implicit **deny-all**.

Why it works. Because evaluation stops at the first match, a broad rule above a specific one shadows it — the specific rule never executes. So order encodes logic: put specific/exception rules above general ones. The implicit deny-all at the bottom means anything not explicitly permitted is dropped — a fail-safe default.

Applied. Rule-ordering PBQ: to allow HTTPS to one server but block all other traffic to its subnet, the "permit tcp/443 to server" rule must sit *above* the "deny ip to subnet" rule. Reverse them and the deny fires first, blocking the traffic you meant to allow. Trace each flow against the rules in order.

Fails when an "allow any" rule drifts to the top during changes — silently permitting everything below it and nullifying the rest of the policy.

Confidence: HIGH.

SPF + DKIM + DMARC as a system [4.5] · Application

What. The three email-authentication records work together: SPF checks the sending server, DKIM checks message integrity/origin via signature, DMARC ties them together with an enforcement policy and reporting.

Why it works. Spoofing works when a receiver can't verify a sender. SPF alone breaks on forwarding; DKIM alone doesn't say what to do on failure. DMARC layers on top, requiring alignment and telling the receiver to quarantine/reject failures — closing the BEC/phishing gap any single record leaves. The system only stops spoofing when all three are deployed and DMARC is set to enforce (not p=none).

Applied. A company still gets spoofed despite SPF + DKIM. Cause: DMARC is at p=none (monitor only). Fix: move to p=quarantine or p=reject. This "policy set to none" detail is a common exam trap.

Confidence: HIGH.

4.6 — Identity & Access Management (IAM)

Factor	"Something you..."	Examples
Knowledge	know	Password, PIN, security question.

Factor	"Something you..."	Examples
Possession	have	Phone/TOTP app, smart card, hardware token.
Inherence	are	Fingerprint, face, iris (biometric).
Location	are (somewhere)	Geolocation / geofencing restriction.
Behavior	do	Typing rhythm, gait, usage pattern.

Why MFA must cross factor categories [4.6] · Application→Analysis

What. MFA requires two or more authentication factors from **independent categories**. A password + a PIN is not MFA — both are "something you know."

Why it works. The security gain comes from independence: an attacker who phishes your password (knowledge) still can't produce your hardware token (possession) or fingerprint (inherence). Two factors from the same category share a failure mode — steal one, likely steal both — so they don't multiply the attacker's difficulty. Crossing categories forces the attacker to compromise two unrelated things at once.

Applied. Scenario: an account with a strong password is still breached by phishing. Adding a second knowledge factor wouldn't help — it's phishable too. Adding a possession factor (TOTP/FIDO2 key) defeats the phish because the attacker lacks the physical device. FIDO2/passkeys further resist phishing by binding to the origin.

Confidence: HIGH.

SSO, federation & PAM [4.6] · Application

What. **SSO** lets one authentication grant access to many apps; **federation** extends trust across organizations via an identity provider (SAML/OAuth/OIDC); **PAM** tightly controls and monitors privileged accounts (vaulting, just-in-time access, session recording).

Why it works. SSO reduces password fatigue and reuse, but concentrates risk — the IdP becomes a high-value target, so it must be hardened and MFA-protected. Federation works by the service provider trusting assertions from a trusted IdP rather than managing its own credentials. PAM exists because privileged accounts are the attacker's ultimate goal; limiting, time-boxing, and recording their use shrinks the damage window.

Applied. SAML flow: user hits an app (SP), is redirected to the corporate IdP, authenticates once with MFA, and returns with a signed assertion granting access — no new password. PAM: an admin checks out a credential just-in-time for a 2-hour window; the session is recorded and the password auto-rotates after.

Confidence: HIGH.

4.7 — Automation & Orchestration

Benefits vs. costs of automation (SOAR) [4.7] · Application→Analysis

What. Automation executes single tasks; orchestration chains tasks across tools into a workflow (e.g., a SOAR playbook). Use cases: user provisioning, ticket enrichment, guardrails, and incident response actions.

Why it works. Automation improves consistency, speed, and scale — a playbook enriches and contains an alert in seconds without human error or fatigue. But the same leverage cuts both ways: complexity and technical debt grow, and an automated workflow becomes a single point of failure. A flawed playbook executes its mistake everywhere, instantly. The trade is reliability-at-scale against concentrated, fast-propagating risk.

Applied. Auto-provisioning: HR system triggers account creation with role-based access on hire and disables it on termination — closing the "orphaned account" gap. But if the joiner-role mapping is wrong, every new hire gets over-privileged at once. So automation needs the same change control and testing as any production code.

Fails when a single automated pipeline has no fallback — its outage or logic bug halts or corrupts every workflow that depends on it (single point of failure).

Confidence: HIGH.

4.8 — Incident Response

Phase	Goal / key activity
1. Preparation	Plans, tools, training, communication trees before anything happens.
2. Detection	Identify the incident from alerts/indicators.
3. Analysis	Scope it — what/where/how bad; validate it's a real incident.
4. Containment	Limit spread (isolate hosts, disable accounts).
5. Eradication	Remove the threat (malware, backdoors, vulnerable config).
6. Recovery	Restore systems to normal, validated, and monitored.
7. Lessons learned	Post-incident review; feed improvements back to preparation.

Why the lifecycle order is fixed [4.8] · Application→Analysis

What. The phases run in sequence, and the sequence is causal — you cannot skip ahead. Supporting activities: tabletop exercises, root-cause analysis, threat hunting, and digital forensics (**legal hold, chain of custody**).

Why it works. Containment must precede eradication because removing malware while the attacker still has an open channel lets them re-enter — you eradicate into a still-open door. Analysis must precede containment because containing the wrong scope either misses infected hosts or needlessly downs clean ones. Each phase creates the precondition the next one depends on, which is why order is the exam's favorite trap.

Applied. Ransomware: detect the mass-encryption alert → analyze scope (which shares/hosts) → contain by isolating affected VLANs and disabling the compromised account → eradicate the malware and backdoors → recover from offline backups → lessons learned updates the playbook. Chain of custody is preserved if evidence may support prosecution.

Fails when responders wipe/re-image before forensic capture — destroying evidence and breaking chain of custody, so the intrusion can't be fully understood or prosecuted.

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Confidence: HIGH.

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1. Preparation	Plans, tools, training, comms trees — before anything happens.
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3. Analysis	Scope it — what/where/how bad; confirm it's real.
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5. Eradication	Remove the threat (malware, backdoors, bad config).
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Fails when responders wipe/re-image before forensic capture — destroying evidence and breaking chain of custody, so the intrusion can't be fully understood or prosecuted.

Confidence: HIGH.

4.9 — Investigative Data Sources

Data source	What it tells you in an investigation
Log data	Sequence of events — auth, firewall, application, system logs.
SIEM dashboards	Correlated cross-source view; alert timelines.
Packet captures	Ground truth of what actually traversed the wire.

Data source	What it tells you in an investigation
Vulnerability scans	Which weaknesses existed on the host at the time.
Automated reports	Trends, recurring alerts, compliance status.
Endpoint/EDR telemetry	Process trees, file changes, persistence artifacts.

Reading logs to reconstruct an attack [4.9] · Analysis

What. Investigation correlates multiple data sources into a timeline that answers what happened, when, and how — the core PBQ skill of this objective.

Why it works. Each source has different fidelity and coverage. Logs are broad but can be tampered with or incomplete; packet captures are authoritative but voluminous and often unavailable retroactively. Corroborating a claim across independent sources raises confidence, while a gap in one source tells you where the attacker may have cleared tracks. The timeline is only as strong as the weakest corroborated link — the evidence-hierarchy logic behind sound conclusions.

Applied. Log PBQ: a firewall log shows an allowed inbound connection to a web server at 02:14; the web log shows a suspicious POST at 02:14; the auth log shows a new admin login at 02:16; EDR shows a new service at 02:17. Read in order, that chain = exploited web app → privilege escalation → persistence. The mitigation targets the entry point (patch the web app), not just the symptom (the rogue service).

Fails when clocks across sources aren't synchronized (no NTP) — the timeline misorders events and the causal chain becomes unreadable.

Confidence: HIGH.

DOMAIN 5 — Security Program Management & Oversight

20%

Governance, risk, and compliance — the business layer that decides what security must achieve and proves it does. Memorization-friendly but heavily weighted; it decides passes. Know the risk math (SLE/ALE/ARO), the vendor-agreement acronyms, and the roles.

5.1 — Security Governance

Element	Definition	Nature
Policy	High-level management intent/rules.	Mandatory, broad.
Standard	Specific mandatory requirements.	Mandatory, specific.
Procedure	Step-by-step how-to.	Mandatory, detailed.
Guideline	Recommended best practice.	Optional, advisory.

Data roles & why they're separated [5.1] · Recall→Application

What. Key roles: **data owner** (accountable, sets classification), **data controller** (decides why/how data is processed), **data processor** (processes on the controller's behalf), **data custodian/steward** (implements the controls day-to-day).

Why it works. Separating accountability (owner/controller) from operation (custodian/processor) enforces separation of duties: the person who decides how data should be protected is not the same person who executes it, which creates a check against both error and abuse. It also maps cleanly to privacy law, where controller and processor carry distinct legal obligations.

Applied. A hospital (controller) sets patient-data handling rules; a cloud billing vendor (processor) only processes as instructed; the IT team (custodian) applies the encryption and access controls; the department head (owner) signs off on the classification. The exam tests which role is accountable vs. which merely executes.

Confidence: HIGH.

5.2 — Risk Management Process

Quantitative risk: SLE, ARO, ALE [5.2] · Application→Analysis

What. Quantitative risk uses money and frequency. **SLE** (Single Loss Expectancy) = Asset Value × Exposure Factor. **ARO** (Annualized Rate of Occurrence) = how many times per year. **ALE** = SLE × ARO — the expected annual loss.

Why it works. ALE converts risk into a dollar figure you can compare against the cost of a control — the core cost/benefit test. A control is economically justified only if it reduces ALE by more than it costs annually. This is why the math matters: it turns "should we buy this?" from opinion into a defensible calculation, and it separates quantitative (objective, data-driven) from qualitative (relative High/Med/Low) assessment.

Applied. A \$100,000 server with a 30% exposure factor per flood → SLE = \$30,000. Floods hit once every 5 years → ARO = 0.2. ALE = \$30,000 × 0.2 = \$6,000/yr. A \$2,000/yr flood control that eliminates the risk is justified (saves \$6k for \$2k); a \$9,000/yr control is not. Expect the exam to give you three of the four values and ask for the fourth.

Confidence: HIGH.

Also tested: **risk register** (log of risks, owners, treatments), **risk tolerance/appetite**, treatment strategies — **mitigate, transfer, accept, avoid** — and BIA metrics **RTO, RPO, MTTR** (mean time to repair) and **MTBF** (mean time between failures).

5.3 — Third-Party Risk

Agreement	Purpose
SLA	Service Level Agreement — measurable performance/uptime commitments.
MOU	Memorandum of Understanding — informal statement of intent, not binding.
MSA	Master Service Agreement — overarching terms for an ongoing relationship.
NDA	Non-Disclosure Agreement — protects confidential information.
BPA	Business Partners Agreement — terms between partners in a venture.

Why vendor risk needs continuous management [5.3] · Application

What. Manage third parties via **due diligence** (vet before onboarding), **right-to-audit** clauses, **supply-chain analysis**, questionnaires, and **ongoing monitoring**.

Why it works. A vendor's vulnerabilities become yours because you extend trust — and access — to them. Point-in-time vetting is insufficient because the vendor's posture drifts after onboarding: they change subprocessors, suffer breaches, or let controls lapse. Continuous monitoring and the contractual right to audit keep the trust relationship tied to current evidence, not a one-time snapshot. The SolarWinds-style supply-chain compromise is the reasoning behind this objective.

Applied. Onboarding a SaaS payroll vendor: run due diligence (SOC 2 report review), embed a right-to-audit and breach-notification clause in the MSA, require an NDA, and schedule annual questionnaires plus continuous security-rating monitoring. If their SOC 2 lapses, the contract lets you audit or exit.

Confidence: HIGH.

5.4 — Security Compliance

Compliance, privacy roles & consequences [5.4] · Recall→Application

What. Compliance = meeting external legal/regulatory/contractual obligations, demonstrated through reporting, monitoring, and **attestation**. Privacy distinguishes the **controller** (decides processing) from the **processor** (acts on instructions) and grants rights like the **right to be forgotten**.

Why it works. Compliance and security overlap but are not the same: compliance proves you meet a defined bar, while security reduces actual risk — you can be compliant and still insecure. The consequences of non-compliance (fines, sanctions, contract loss, reputational damage) are what make compliance a business driver, giving the security program its funding and mandate. Privacy roles matter because liability follows the role.

Applied. Under GDPR, a company (controller) using a cloud analytics vendor (processor) must document the lawful basis for processing and honor a customer's deletion request across both. Non-compliance risks fines up to 4% of global revenue — the "consequences" the exam links to executive buy-in.

Confidence: MODERATE — specific thresholds are regulation- and jurisdiction-dependent; verify current figures.

5.5 — Audits & Assessments

Type	Meaning
Internal audit	Self-review by the org; continuous improvement, pre-check for external.
External audit	Independent third party; objectivity and regulatory credibility.
Attestation	Formal written declaration of audit findings.
Pen test — known (white-box)	Tester has full internal knowledge; deep, efficient.
Pen test — partially known (gray)	Some knowledge; balances realism and coverage.
Pen test — unknown (black-box)	No prior knowledge; simulates a real external attacker.

Internal vs. external, and pen-test knowledge levels [5.5] · Application→Analysis

What. Audits verify controls exist and work; assessments identify gaps. Pen tests actively exploit them. Testing environments range from known (full info) to unknown (none), and reconnaissance precedes the test.

Why it works. The choice trades realism against efficiency. Black-box testing mirrors a real attacker's ignorance and tests detection, but wastes budget rediscovering what you already know; white-box skips reconnaissance to probe depth but doesn't test whether your team would notice an unknown intruder. External audits add objectivity that internal ones structurally cannot — you can't fully grade your own work — which is why regulators require them. Matching method to the question being asked is the reasoning tested.

Applied. To satisfy a regulator, commission an external audit for objectivity. To test whether the SOC detects a novel intrusion, run a black-box pen test. To thoroughly harden a specific new application before launch, a white-box test is the efficient choice.

Confidence: HIGH.

5.6 — Security Awareness

Awareness as a security control [5.6] · Application

What. A program of phishing simulations, anomalous-behavior recognition training, user education, and clear reporting channels, executed and measured continuously.

Why it works. Users are the most-targeted attack surface (social engineering) and the one technology can't patch, so training is a genuine preventive/detective control, not a compliance checkbox. It works by changing the base rate: a trained user is less likely to click and more likely to report, and each report is an early detection signal. Measuring click and report rates turns "awareness" into a metric you can improve — the input→action→measurement→feedback loop.

Applied. A monthly phishing simulation drops the click rate from 18% to 4% over two quarters while the report rate rises — early-warning coverage improves measurably. A spike in click rate after a template change flags where to focus the next training cycle.

Confidence: HIGH.

Exam-Day Strategy & Recap

Use these notes as a reasoning trainer, not a glossary. For each concept you can already define, force yourself to answer the *Why it works* and *Fails when* layers out loud — that is the depth the scenario and performance-based questions reward.

Priorities

Priority	Why
Master Domain 4 (28%)	Largest weight and where PBQs cluster — firewall ordering, log analysis, tool matching.
Don't defer Domain 5 (20%)	GRC is memorization-friendly and decides passes; the risk math (ALE) is easy points.
Drill PBQ patterns	Device placement (3.2), rule ordering (4.5), log reading (4.9), control matching (1.1).
Reason, don't memorize	Distractors target rote learners; the causal "why" disambiguates close answers.

Logistics recap: up to 90 questions in 90 minutes — about a minute each, so flag and move on. Passing score is 750/900. Read for **bold**/*italic*/**UPPERCASE** qualifiers and choose the *best* answer, not merely a correct one.

Grounded in the SecPlus SY0-701 knowledge base and the official SY0-701 (V7), objectives v5.0 blueprint. Practice content is original and follows CompTIA's authorized-materials policy. Confidence tags reflect evidence weight; [VERSION-SENSITIVE] items should be re-verified against current CompTIA objectives.